

After the Flood—There's Hope

Homes that are damaged by flooding can be repaired and made more durable.

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Flooding is a fact of life that results in more damage to buildings throughout the United States than any other single natural cause. Residential buildings are especially hard hit during floods; the lives of the residents are disrupted and precious possessions are lost. While homeowners are discouraged or prohibited from building new homes in flood-prone areas, existing homes, when damaged, must be repaired. Builders can minimize damage to these homes from future flooding by repairing or renovating them using materials and methods that are resistant to flood damage.

Researchers at Oak Ridge National Laboratory (ORNL) and Tuskegee University have been working to identify and evaluate building materials, systems, and methods that can be used to repair houses after flooding and that make the envelope of a house more resistant to future flood damage (see "After the Flood," *HE* Jan/Feb '03, p. 12). We were also interested in providing a scientific basis for certifying building materials and systems that are resistant to flood damage, so that that information can inform future building code development.

The Department of Homeland Security, Emergency Preparedness and Response Directorate defines flood damage resistance as the ability of materials, components, and systems to withstand direct and prolonged contact with floodwater without sustaining degradation that requires more than cosmetic repair to restore it to its original condition. Cosmetic repair includes cleaning, sanitizing, and resurfacing material; resurfacing includes sanding, repairing joints, and repainting. Repair should be less costly than simply replacing the affected materials and systems. As a



The first crawlspace test module tested hardboard lap siding, plywood sheathing, standard wood framing, fiberglass batt insulation, gypsum drywall, plywood subflooring, and a variety of interior finishes.

result of our materials testing, we have identified another key attribute of flood damage resistance. Individual materials that are considered flood damage resistant must not cause the degradation of adjacent materials, or the systems of which the material is a part.

A 2-ft Flood

Our test facility is located on the experimental farm near an agricultural lake at Tuskegee University, in Tuskegee, Alabama. Initially, two test modules were built with typical home construction materials and methods. The modules were flooded to a level of 2 ft above the floor with water from an agricultural lake that reasonably represented typical riverine floodwater. The test modules were allowed to dry for 30 days. Detailed information was col-

lected to determine how these construction materials and systems were affected during and after flooding. Test results from these materials and systems provided a baseline against which other materials and systems could be compared.

Based on what we learned in those tests, we conducted a second set of tests. The second set of tests introduced materials and systems that we expected to be more flood damage resistant than typical construction materials and systems. In the second tests, unlike the first ones, we sanitized the materials and systems while they were drying, and cosmetically restored them after the drying period, in order to assess their performance after exposure to a flood. Samples of the various materials were taken for testing and observation, and the test modules were then demolished and autopsied. The ini-

Instrument Locations

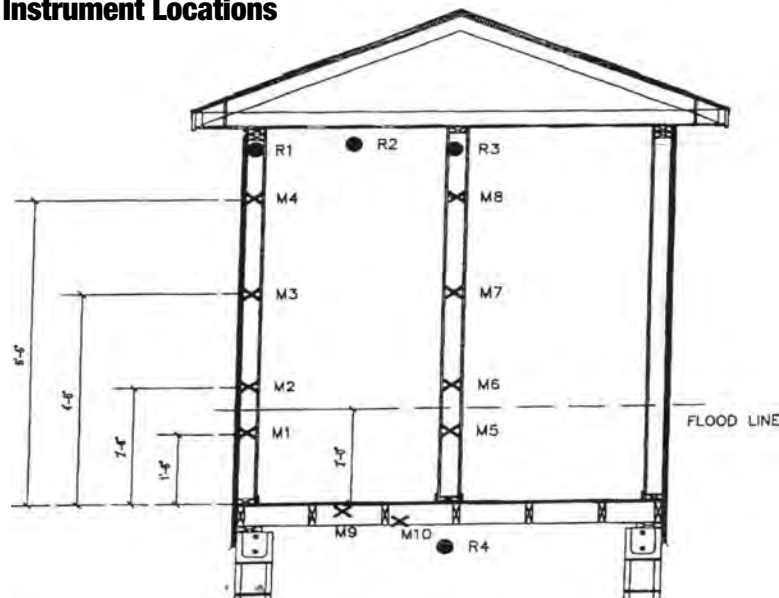


Figure 1. Moisture sensors (M) were located in studs in the exterior wall (left), interior partition (center), in the underside of the subfloor, and in a floor joist. Relative humidity sensors (R) were located near the ceiling in one room, at the top of two walls, and, after draining the floodwater, in the crawlspace.



Spray polyurethane foam (SPUF) insulation was tested several times. It was unaffected by exposure to water and permitted adjacent materials to dry to pre-flood levels within the test period. The picture was taken during the autopsy of the module.

tial results of our research were reported in the article cited above.

We built a third pair of modules to attempt dry floodproofing—constructing a building in such a way that no water enters the structure during flooding—and to test additional materials and systems for resistance to flood damage.

In all of our testing, we employed relative humidity transmitters; thermocouples that measured temperature; and moisture sensors installed in wall studs, wall surfaces, floor joists, and floor surfaces (see Figure 1). A weather station provided data on ambient conditions during the test. We used a hand-held

moisture meter to measure material moisture content during the postflood drying period. Mold was sampled from the modules and tested in a laboratory to identify its type. Flexural strength and modulus were determined for various types of siding and wallboard.

We also developed detailed protocols for visual observation. While visual observation is subjective, the protocols were developed to systematize these observations and make them as detailed and consistent as possible throughout the series of tests. We made extensive photographic and video records as well.

Flood-Damage-Resistant Materials and Systems

The ORNL/Tuskegee experiments tested only for resistance to physical degradation that results from the wetting and drying cycle associated with flooding. The testing did not address the structural impact on the envelope of externally applied hydrostatic pressures. Postflooding mold growth was documented and selected specimens were analyzed. Some test modules were also cleaned and sanitized to determine if mold growth could be controlled. However, bacteriological and toxic materials testing were not performed during this series of tests. And we did not test for the residual health effects of flooding on otherwise flood-damage-resistant materials and systems. Such testing could potentially change our conclusions.

The following conclusions should be viewed as preliminary, since the purpose of our project was to provide a scientific basis for certifying materials and systems for flood damage resistance for use in building codes, and as yet there is no accepted certifying test procedure.

Siding

Newly installed and painted plywood and hardboard lap siding maintained reasonable dimensional stability and mechanical properties after they were dried. The siding materials that we tested were washable after flooding and drying, but these materials remained discolored. However, older, weathered plywood and hardwood siding, or siding made of the same materials that is repeatedly wetted

and dried over several cycles, could significantly degrade and require more than cosmetic repair to restore it to its original condition. Both vinyl and fiber cement siding could be restored to pre-flood conditions through washing the portion below flood level. The corner boards, which were made from nominal 1-inch sawn lumber, cracked and warped. Vinyl corner trim showed no evidence of deterioration from flooding.

Sheathing

Fiberock sheathing, a water-resistant, fiber-reinforced gypsum panel manufactured by USG, maintained its integrity and mechanical properties. The Fiberock dried to preflood conditions during the drying period.

Plywood sheathing maintained its integrity and mechanical properties. However, it had not dried to preflood conditions after 30 days. Because water does not tend to escape quickly from behind plywood siding, we do not consider the combination of plywood siding and sheathing to be a good flood-damage-resistant system. Lap siding tends to let moisture escape more quickly. If a flood-damage-resistant lap siding is employed, the use of plywood as sheathing is likely to make an acceptable flood-damage-resistant system.

We experimented with 15# felt under both vinyl and fiber cement lap siding and found that the sheathing had dried to preflood conditions at the time of autopsy. We will perform a test using Tyvek house wrap at press time. According to the manufacturer, Tyvek inhibits the passage of liquid water and it breathes, so water vapor from drying can pass through the material. We'll know experimentally how it performs under flood conditions in a couple of months.

Wood Framing

The moisture content in wood studs above the flood line returned to preflood levels within the drying period. The portion of the studs below the flood line did not, in most cases, dry to preflood levels during the drying period. We considered wood studs to be flood damage resistant as long as the wall system will permit them to continue to dry to normal moisture levels.



This is an inside view (taken at autopsy) of an open-cavity wall showing the back of a Fiberock gypsum panel after exposure to flooding. Note that there is no visible evidence of mold growth. The dark spots on the second stud from the right are knots or pitch pockets.



When we reentered the test module eight days after flooding, we found mold growth above the waterline and staining below it. Mold growth increased and the staining remained after we washed the test module ten days after flooding.

Wall Insulation

Fiberglass batt insulation appeared to retain moisture in the exterior wall cavities and below the floor (see Figure 2). The moisture on the fiberglass fibers appeared to keep adjacent walls and floor materials wetter longer; this could potentially cause long-term damage to the subflooring, to the floor and wall framing, and to the gypsum board walls. When spray polyurethane foam (SPUF) insulation was used in the wall cavities, the wallboard and wood studs in exterior walls dried at nearly the same rate as in the interior walls with empty cavities (see Figure 3, p. 22). SPUF absorbs water very slowly and was undamaged by flooding. SPUF did not retain moisture; thus it did not reduce the flood damage resistance of the materials around it.

Interior Wallboard

When conventional paper-faced gypsum board was used with fiberglass batt insulation on exterior walls, the gypsum board lost about 50% of its flexural strength; at the end of the drying period, it remained wetter than the gypsum board on uninsulated interior walls. Interior gypsum board walls dried out and regained flexural strength during the test period. The gypsum board could be restored to preflood conditions with cosmetic restoration (see the article cited above for details).

Conventional drywall supported mold growth during several tests, but it was cleaned and sanitized, which eliminated the mold stains. Fiberock interior wallboard, a non-water-resistant gypsum product without paper facings by USG, regained about 70%–78% of its initial strength when tested (see Table 1), and it dried out during the 30-day drying period. Although it supported mold growth, it could be cleaned, sanitized, and restored.

Fiberock exterior sheathing was applied to some interior walls. It regained 82%–87% of its initial strength on an exterior wall insulated with SPUF and 101%–130% of its initial strength on an interior, hollow-cavity wall. (The strength actually increased in one instance after exposure to flooding. This happens occasionally with gypsum products, since complete hydration of all of the gypsum does not always occur in manufacturing, and subsequent exposure to water completes the process.) The Fiberock exterior sheathing applied to interior walls also dried out during the drying period. It did not support mold growth and its surfaces were easily cleaned and restored.

Wall Finishes

Ceramic tile performed well under flood conditions and showed no long-term deterioration. Both latex flat paint and latex semigloss enamel paint peeled, blistered, and stained. Mold grew on both types of paint. High- and low-permeability paints were tested. Both types of paint had to be sanded and new coats of paint applied to restore the walls to preflood conditions. We also compared water-based flat latex and oil-based enamel paint. The water-based latex flaked and

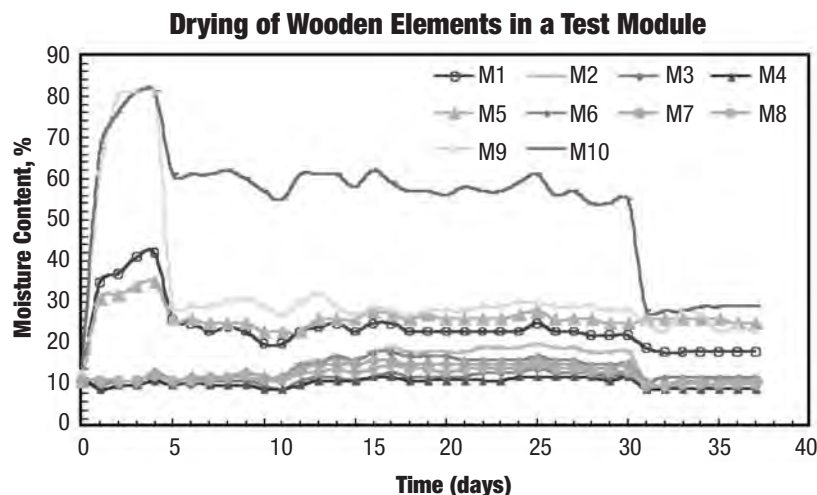


Figure 2. This figure indicates the moisture content in wooden elements of the first crawl space test module. Note the high moisture level (55%-60%) being shown by sensor M9, which is located on the underside of the plywood subflooring above 6-inch fiberglass batt flooring insulation. On day 31 (end of the drying period), the batts were removed and the moisture level dropped to approximately 30%.

Table 1. Results of Mechanical Properties Testing

Location	Thickness In (mm)	Observation Period	Mechanical Properties	
			Ultimate Flexural Strength* (MPa)	Modulus (GPa)
Interior: Fiberock Interior Wallboard				
As received	0.5 (12.7)	Before flood	3.99	2.06
Interior wall above water	0.5 (12.7)	After autopsy	3.55	1.72
Interior wall below water	0.5 (12.7)	After autopsy	2.79	1.34
Interior: Fiberock Exterior Sheathing				
As received	0.5 (12.7)	Before flood	6.19	3.50
Exterior wall above water	0.5 (12.7)	After autopsy	5.83	1.74
Exterior wall below water	0.5 (12.7)	After autopsy	5.09	1.94
Interior wall above water	0.5 (12.7)	After autopsy	4.79	1.77
Interior wall below water	0.5 (12.7)	After autopsy	6.25 [†]	2.12

* The ultimate flexural strength is the better indicator of strength.

[†] In this case the strength actually increased after exposure to flooding. This happens occasionally with gypsum products, since complete hydration of all of the gypsum does not always occur in manufacturing, and subsequent exposure to water completes the process.

blistered. Oil-based flat enamel paint performed better than any other paint that we tested. It flaked and blistered very little and was much easier to restore than other paints. Of all the paints tested, oil-based flat enamel paint was found to be the most flood damage resistant. However, we did not completely investigate the impact of oil-based enamel on the drying of adjacent materials and systems in this testing. Vinyl wall covering blistered, peeled,

and debonded after flooding. It damaged the surface of the gypsum board, and it may inhibit the drying of the substrate or wall system.

Exterior Doors

Exterior wood-paneled doors in a wood frame, and exterior prehung metal-clad doors in a wood frame were stained slightly, but could be washed and restored. Foam-filled fiberglass and

foam-filled metal were restored to pre-flood conditions with minimal effort. The fiberglass and metal doors used in the second modules were reused in the third modules and once again were easily restored.

Interior Doors

All the interior doors that we tested were severely stained after flooding and drying, and some warped, split, and peeled. Considering the relatively low cost of replacement, we did not consider it economically feasible to restore any of these doors.

Windows

All vinyl and aluminum window frames could be restored to pre-flood conditions with minimal effort.

Floor Structure

The sealed-concrete floor slab in all the slab-on-grade modules remained undamaged during and after flooding. The wood subflooring retained very high moisture content throughout the drying period when unfaced fiberglass batt insulation was installed underneath the subflooring. (Two 8 inch x 18 inch foundation vents were installed on opposite walls in the test modules. These were open throughout the drying period. The area of these vents exceeded code minimum requirements for a structure of this size, 64 ft².) When no floor insulation was used, the subflooring returned to pre-flood moisture levels during the drying period. Wood subflooring and framing insulated with fiberglass batts could experience long-term moisture-related problems. At press time another module was being tested using spray polyurethane insulation under the subflooring.

Floor Finishes

Ceramic tile and quarry tile performed very well under flooding conditions and required only cleaning to be restored. All carpeting and padding became dirty and smelly after flooding. It also retained large amounts of moisture, which would slow the overall drying rate throughout the house. Even if the carpet is able to withstand the flood, it should be removed for cleaning and drying, and to promote drying within the home.

Simulated wood flooring, a composite wood fiber and plastic material, warped and had open joints when it was left in place after flooding. This flooring had much less warping and shrinkage when it was removed, washed, and stacked to dry after flooding, but the process of removal damaged some of the pieces.

Foundation Vents

The operable flood vents (Smart Vent) were closed prior to flooding and opened by themselves during the filling and draining of the floodwater. They operated as designed. These vents were blocked open throughout the drying period. The crawlspace humidity reached 100% and remained high during the drying period. This humidity level is unacceptable in the long term, since it could contribute to both mold and wood decay. We believe that the high humidity level in the crawlspace resulted from the test module being placed in a basin that collected a significant amount of rain-water throughout the drying period. In order to keep it from providing a path for mold to enter the house, the crawlspace area must be effectively sealed from the interior of the building.

Procedures for Responding to Flood Damage

Along with evaluating materials for their flood-damage-resistant properties, we tested the effectiveness of some procedures for dealing with flood damage.

Punching Holes in Walls

Punching holes above the floor molding of the interior walls does not drain any water, nor does it dry the wall any faster, especially if floodwater has receded for several hours. In some instances, if holes are not punched in the walls, the gypsum board can easily be repaired and restored. Punching holes in gypsum board walls to promote drainage is not an appropriate flood recovery procedure.

Cleaning

The cleaning protocols that we followed used a clear-water rinse that did

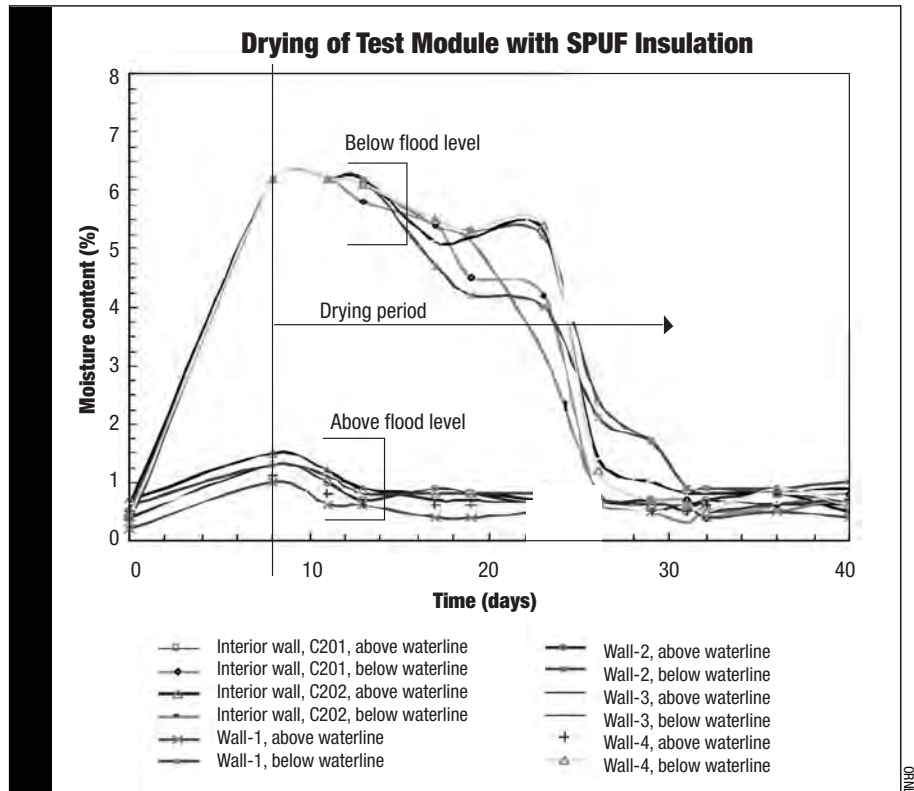


Figure 3. All gypsum wall products returned to preflood moisture levels during the test period (ending on day 31) when SPUF was used as wall insulation.



After the floodwater had penetrated the outer barriers, it entered the interior through the joint between the sill plate and the studs, as well as between the studs where they had been doubled (see tools stuck in these joints). The SPUF insulation was effective at preventing water from entering through the cavity of the wall.



Our second try at dry floodproofing involved removing the vinyl siding below flood level, caulking all visible cracks or holes in the sheathing and around the window, caulking the sill plate joint with the slab, and covering that joint with butyl rubber tape. A window dam made from 2-inch-thick polystyrene insulation was fabricated and installed in the window frame. The dam was held in place with silicon caulk. Despite all these efforts, the envelope leaked when subjected to flooding, albeit at a slower rate than an unprotected structure.

remove some dirt and staining, but not mold. A second washing with soap and water on selected materials (vinyl and fiber cement siding, fiberglass doors and window frames) did restore them to preflood conditions. After sanitizing with a solution of bleach (trisodium phosphate) and water, most elements that were not physically damaged could be restored to preflood conditions.

Sanitizing

Severe mold growth occurred in the first tests when no attempt was made to clean or sanitize surfaces during the drying period. Mold growth also occurred on exposed interior surfaces in most subsequent tests, and efforts were made to sanitize surfaces and remove mold. After sanitization, there was no visible mold for the remainder of the testing period. Although no mold reappeared throughout the test period, we cannot verify the long-term elimination of mold.

After demolition and autopsy of these units, it was determined that there was very little or no mold growth in the nonexposed (hidden) portions of the structure that were not sanitized. Sani-

tizing appeared to work on the exposed surfaces of the modules to eliminate mold growth, and it is therefore a recommended procedure in the restoration of flood-damaged homes. Sanitizing the nonexposed portions of the structure for mold control does not appear warranted, based on what we saw during the autopsy.

Restoring

All exterior siding tested could be restored, as could interior wallboards when SPUF was the insulation used. Ceramic tile floors and the sealed-concrete slab were easily restored. All windows and exterior doors tested could be restored.

Dry Floodproofing

We did not achieve dry floodproofing in two attempts (see photo). While the door and window dams that were designed and built were effective in preventing the entry of water through doors and windows, water entered the units through other paths, such as the joint between the interior partition and the exterior walls at floor level. Although the joint between the sill plate and the con-

crete slab had been caulked, water entered there as well. Additional steps were taken in the second attempt; the external joint between the sill and the slab and other potential leak pathways on the exterior were sealed. Despite these efforts, floodwater entered the modules. Based on these failures, dry floodproofing is not considered an appropriate approach to flood damage resistance.

More Work Is Needed

The primary purpose of our project was to identify and evaluate materials, systems, and methods that will make the envelope of a house more resistant to flood damage. These materials, systems, and methods are primarily intended for use to repair houses subsequent to flooding. Then if the house floods again, damage will be reduced and restoration costs and efforts will be minimized.

Besides identifying flood-resistant materials, the purpose of our project was to develop a method to collect representative, measured, and reproducible data on how various materials and systems respond to flooding conditions, so that future recommendations for repairing flood-damaged houses can be based on scientific data. This methodology can then be used to develop a standard test procedure to certify which building materials and systems are resistant to flood damage; the use of these materials and systems for repairing and rebuilding flood-damaged homes in flood-prone areas could be required by future building codes.

A certifying test procedure must be developed and adopted by a certifying agency before the identification of materials as “flood damage resistant” will satisfy the requirement for the use of such materials by a building code group or local jurisdiction. This project, and other related activities at ORNL and Tuskegee, are contributing to the development of that certifying test procedure.



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